# ADITYA UNIVERSITY

**Department of freshman Engineering**

**Important Questions**

**Course name:** Basic Electrical and Electronic Engineering **Course code:** 2501EE01

| UNIT-3 | | | | | |
|---|---|---|---|---|---|
| **QNO** | **Sub QNO** | **Question** | **BT** | **CO** | **Marks** |
| 1 | a | Explain the layout diagram of Thermal Power Plant and Describe each part in detail. | L3 | CO3 | 4M |
| | b | A factory has the following appliances and they are in operation:<br>2 HP motor – single quantity, operates 3 hours per day<br>100 W lamp – two quantities, operates 12 hours per day<br>1000 W heater – single quantity, operates 3 hours per day<br>120 W fan – two quantities, operates 6 hours per day<br>Calculate the total energy consumption in a month of 30 days,per unit cost 3 rupees and also calculate the energy bill in a month. | L2 | CO3 | 4M |
| 2 | a | Draw the Hydro electric power plant layout and Explain all its components. | L3 | CO3 | 4M |
| | b | What is Eletctric shock.Explain the safety prication to avoid electric shock. | L2 | CO3 | 4M |
| 3 | a | Draw the layout diagram of a wind power plant and Explain each part. | L3 | CO3 | 4M |
| | b | Explain the operation of fuse and MCB. | L2 | CO3 | 4M |
| 4 | a | Draw the layout diagram of solar power plant and Explain each and every component. | L3 | CO3 | 4M |
| | b | Explain different types Earthing. | L2 | CO3 | 4M |
| **UNIT-4** | | | | | |
| **QNO** | **Sub QNO** | **Question** | **BT** | **CO** | **Marks** |
| 1 | a | Illustrate the operation of P-N junction diode and draw V-I characteristics with neat circuit diagram. | L2 | CO4 | 4M |
| | b | What is transistor biasing? Explain the need for biasing in BJT. | L3 | CO4 | 4M |
| 2 | a | Compare P–N junction diode and Zener diode. | L2 | CO4 | 4M |
| | b | Illustrate common emitter mode of transistor & draw input and output characteristics. | L3 | CO4 | 4M |
| 3 | a | How Zener diode acts as a Zener voltage regulator and explain with neat circuit diagram. | L2 | CO4 | 4M |

| | | | | | |
|---|---|---|---|---|---|
| | b | Describe common base mode of transistor & draw input, output characteristics. | L3 | CO4 | 4M |
| 4 | a | Define cut-in (knee) voltage and breakdown voltage of a P–N junction diode. | L2 | CO4 | 4M |
| | b | Describe the operation of BJT(NPN or PNP transistor). | L3 | CO4 | 4M |
| 5 | a | Differentiate between Zener breakdown and Avalanche breakdown. | L1 | CO4 | 4M |
| | b | Compare CE, CB, and CC configurations of a BJT. | L3 | CO4 | 4M |
| 6 | a | Define static resistance and dynamic (AC) resistance of a P–N junction diode. | L2 | CO4 | 4M |
| | b | Explain the difference between NPN and PNP transistors and Explain the operation of NPN transistor. | L3 | CO4 | 4M |

**UNIT-5**

| QNO | Sub QNO | Question | BT | CO | Marks |
|---|---|---|---|---|---|
| 1 | a | What is Adder? Explain half adder & full adder and draw the corresponding truth table & logic diagram. | L2 | CO5 | 4M |
| | b | Explain L-filter & C-filter and draw the corresponding wave forms. | L3 | CO5 | 4M |
| 2 | a | Explain the operation of full wave rectifier & draw wave forms & derive the expression for average output voltage and current. | L3 | CO5 | 4M |
| | b | Encode a binary word 1100110101 into odd parity hamming code. | L3 | CO5 | 4M |
| 3 | a | Explain the DC regulated power supply with neat diagram. | L2 | CO5 | 4M |
| | b | Explain the logic gates with truth tables . | L3 | CO5 | 4M |
| 4 | a | Explain the Bridge rectifier with neat diagram and waveforms | L2 | CO5 | 4M |
| | b | Encode the given number into odd parity hamming code 1101100. | L3 | CO5 | 4M |
| 5 | a | Explain the i)Half wave rectifier ii) full wave rectifier with neat diagrams. | L2 | CO5 | 4M |
| | b | Explain different codes a. Hamming code b. Grey code c. Xs-3 code | L3 | CO5 | 4M |
| 6 | a | Illustrate the operation of Full wave Centre tapped rectifier & draw the wave forms. | L3 | CO5 | 4M |
| | b | Discuss the operation of NAND, NOR, XOR & XNOR with truth table. | L2 | CO5 | 4M |